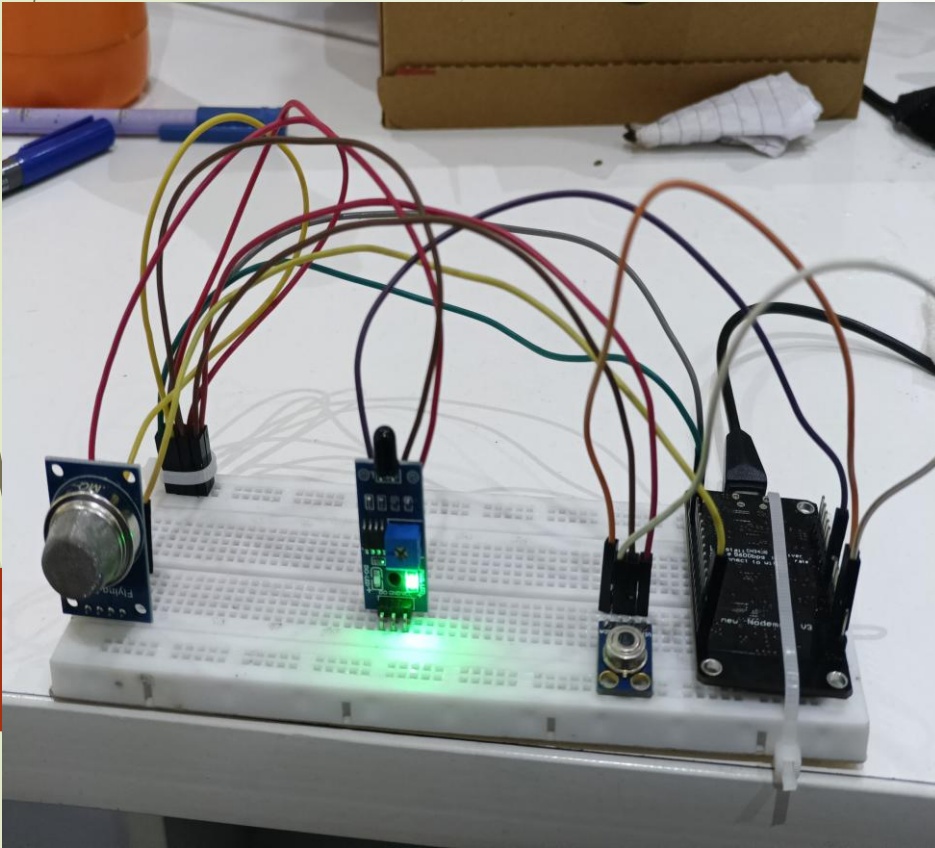


Project Title:
Smoke Detection Using Arduino

Team details:

- ❖ Team Debug Dwellers
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Project Introduction



- **Unique Features:**

- Fully automated diagnostics of fire safety systems with self-check mechanisms.
- Real-time cloud-based monitoring and AI-driven anomaly detection for predictive maintenance.
- Integrated user-friendly dashboard for centralized control, alerts, and compliance tracking.

- **Innovation:**

- Combines IoT, cloud computing, to transform traditional fire safety management.
- Uses automatic system to predict failures, reducing downtime and enhancing reliability.

- **Comparative Advantage:**

- Reduces human interfere, minimizing errors and inspection costs compared to manual systems.
- Offers scalable and flexible implementation for diverse facilities, unlike rigid systems.
- Provides regulatory compliance with automated reporting and real-time insights.

Methodology

Phase 1:

Problem Analysis and Requirement Gathering

- Analyze issues in existing fire safety systems and gather requirements.
- Define hardware, software, and cloud integration specifications.

Phase 2:

Design and Development

- Design circuits and develop diagnostic and monitoring software.
- Configure dashboard for anomaly detection and predictive maintenance.

Phase 3:

Implementation and Testing

- Integrate hardware and software, and test functionality in a controlled setup.
- Validate sensor accuracy, data transmission, and overall system reliability.

Phase 4:

Final Deployment and Evaluation

- Deploy the system in real-world environments and monitor performance.
- Evaluate results, collect user feedback, and plan for scalability.

Core Components of the System

- **Hardware Components: -**

- Fire Detection Sensors: Smoke detectors, heat sensors, and flame sensors.
- Sprinkler System: Equipped with flow and pressure sensors.
- Control Unit: Arduino , esp8266 or similar hardware for integration and processing.
- IoT Devices: For real-time data transmission and device communication.

- **Software Components:-**

- Diagnostic Software: To run checks on all devices (using MIT app inventor)
- Cloud Integration: Stores performance logs and sends alerts in case of malfunction.
- Dashboard: A user-friendly interface to view system status, logs, and alerts.

- **Communication Protocols:-**

- Wi-Fi or Bluetooth for data transmission.
- access online services to make HTTP requests or save data to the cloud

How It Works

- **Automated Diagnostics:-**

- At scheduled intervals or on demand, the system sends test signals to all fire safety components.
- Sensors check for functionality (e.g., smoke detection sensitivity, sprinkler valve operation).
- Self-check mechanisms ensure no human intervention is needed.

- **Data Collection:-**

- Each device sends its performance data (e.g., voltage levels, response times) to the control unit.
- The control unit aggregates the data and transmits it to the cloud.

- **Cloud Processing:-**

- The cloud analyses data with the pre uploaded parameters for safety .
- Generates real-time alerts for any malfunction or performance degradation.

- **User Interaction:-**

- Alerts are sent to facility managers via SMS, email, or a mobile app.
- A dashboard provides insights into system health and maintenance schedules.

Implementation Plan

Prototype Development:-

- Use Arduino for initial testing with sensors and sprinklers.

Software Design:-

- Build application to connect web server and to send and receive reports of quality checks

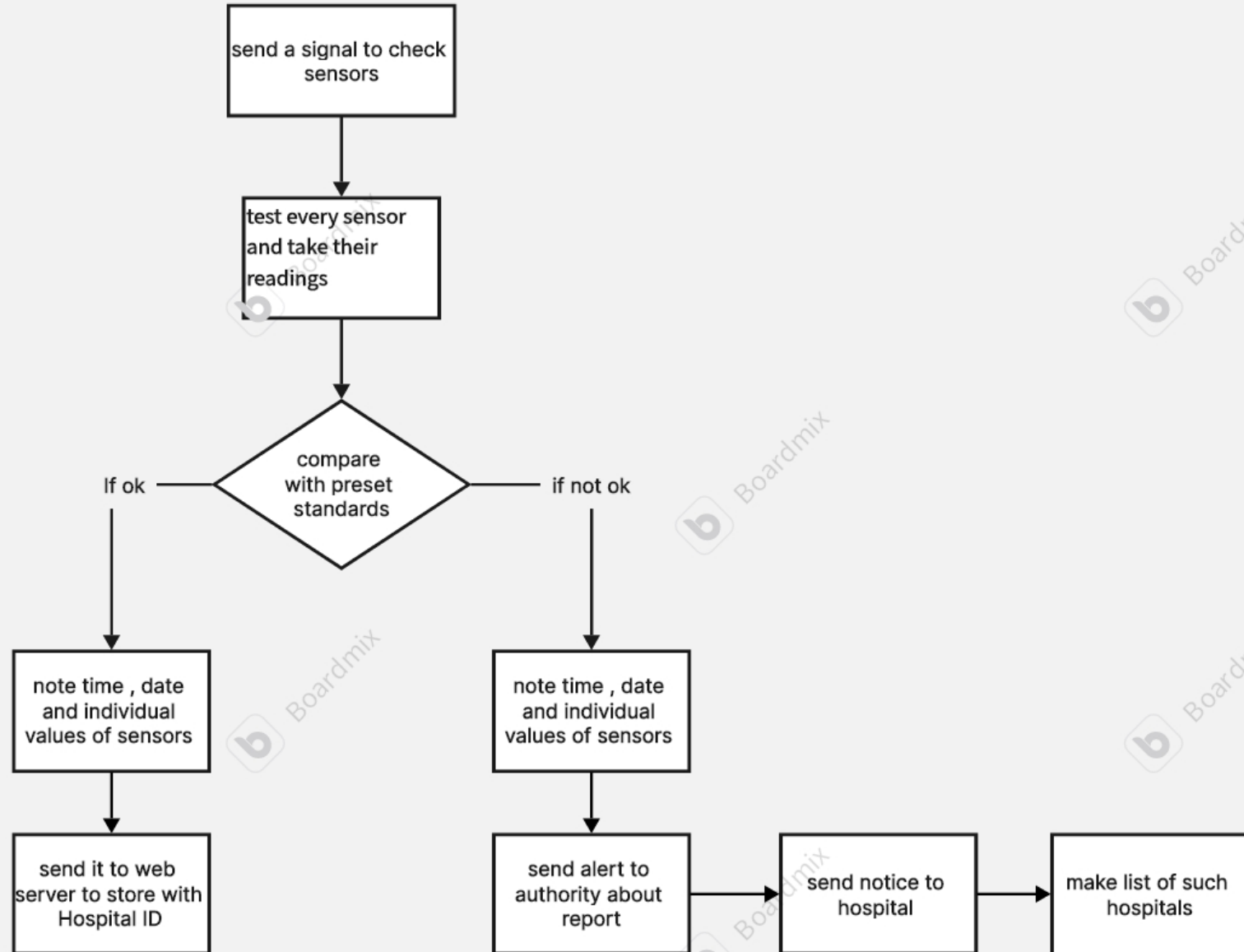
Pilot Testing:-

- Deploy the system in a small hospital or facility for testing.

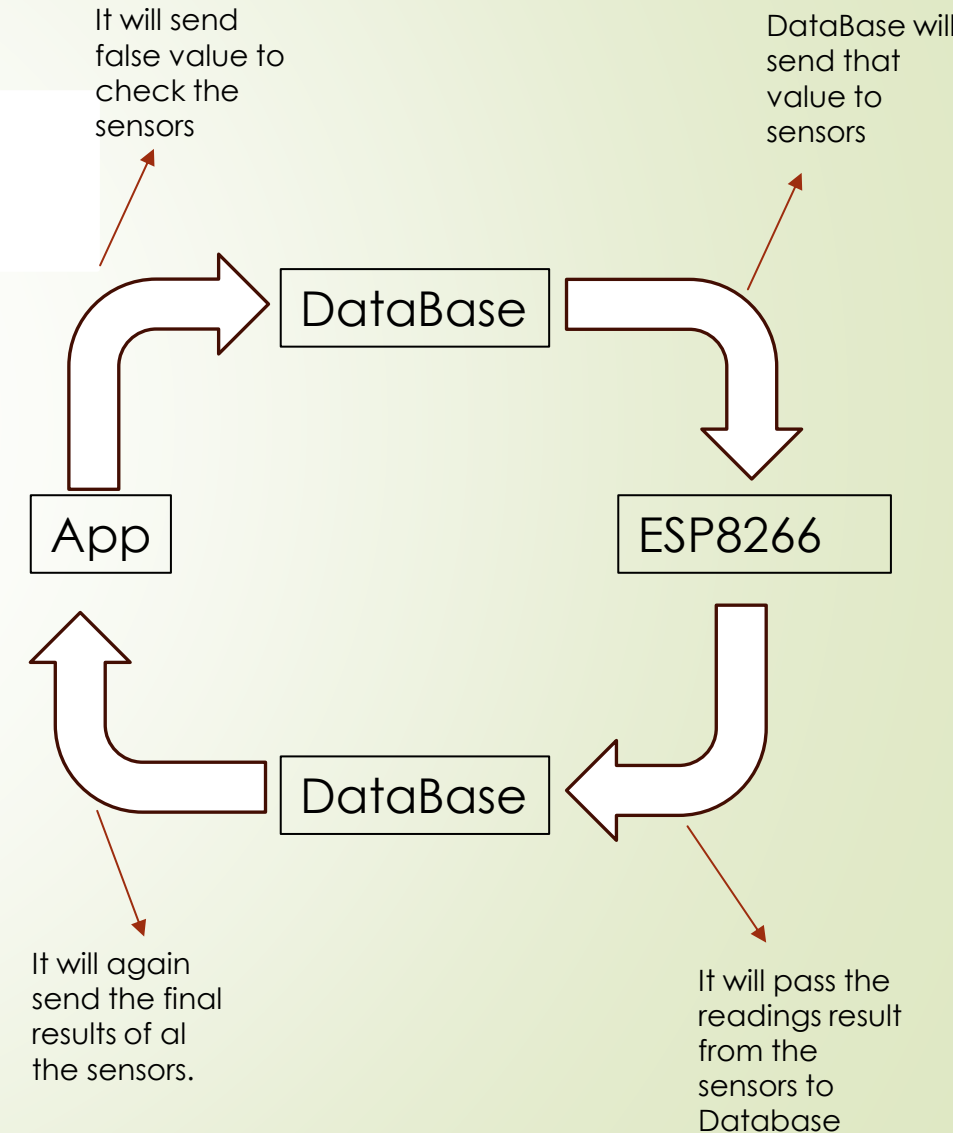
Scaling:-

- Integrate the solution with large-scale facilities and third-party fire safety systems.

Block diagram



CYCLE



Conclusion

This system projects a smart fire safety by incorporating modern technology such as IoT, cloud computing, and AI to enhance safety measures. It reduces human efforts automatically so that all components, from the sensors to sprinklers, function well. Realtime monitoring enables immediate alerts in the event of something going wrong, and predictive maintenance also anticipates problems before they ever occur.

The system is also easy to scale, hence it can be used in places such as hospitals, offices, and even schools. It ensures adherence to safety regulation while saving one time and cost on manual checks. The overall project presents a means of protecting people and their property from hazards by fire in a dependable, efficient, and modern method. The innovation may usher in a new standard for pro-active fire safety management of lives, saving lives while ensuring adherence in critical infrastructures like hospitals.